The Influence of Parental Involvement on Student Engagement in Mathematics

Background of the Study

Student engagement in mathematics is critical, and developing an appreciation for learning is shaped by external factors like parental involvement. When parents actively participate, the classroom environment becomes more supportive, fostering positive attitudes toward math. Parental involvement reassures students, making them more comfortable with the idea of learning math as a collaborative effort. Additionally, math anxiety can hinder student engagement, with the subject often perceived as difficult. Parents can help alleviate this anxiety by providing encouragement and a supportive home environment, boosting students' enthusiasm for learning math.

In Kenya, Performance in Mathematics has remained lower than other subjects in the 8-4-4 system of education in the country, they established that performance in mathematics was lower than other subjects. Furthermore, the Resource Framework Theory by Brooks-Gunn (1995), “human capital” arguments suggest that poor children suffer because of the poorer endowments and investments they receive from their parents (Becker, 1998). Therefore, the Parental involvement is an important variable in determining students engagement in mathematics.

In the Philippines, Marpa (2016) identified attitudes toward mathematics and observed, some students are not performing well in Mathematics because they have not developed the given skills. Problem solving and critical thinking are less developed because they entail students to go beyond their thinking. It suggests the researchers

that to improve performance in mathematics, it is essential to foster these skills through targeted educational practices and support.

The researchers have observed that some students in PCSHS have less engagement in mathematics. This indicates that the teaching method may not be suitable to students, especially to those students who have a lesser critical thinking abilities than others. Parental involvement may play a vital role in student's engagement to mathematics by encouraging them to be more active in it.

While parental involvement is crucial for student engagement in mathematics, there is limited understanding of its specific impact across different educational contexts. In Kenya, most research focuses on socio-economic factors rather than the role of parental involvement in improving mathematics performance. In the Philippines, studies emphasize the development of problem-solving skills but provide little insight into how parents can support this. Additionally, there is a gap in research on the relationship between teaching methods, parental involvement, and student engagement in schools like PCSHS, where students' critical thinking abilities differ.

Statement of the Problem

The purpose of this study is to look into how parental involvement affects students' interest in mathematics. It specifically aims to ascertain whether parents' active involvement in their children's education has a favorable effect on their motivation and academic achievement. The study will also investigate how students' attitudes and

anxieties around learning mathematics are influenced by different parental support levels.

Research Questions

1. How does parental involvement affect the student's engagement with the mathematics subject?

1.1 Parents expectations

1.2 Parents- child communication

1.3 Homework support

1.4 School-based involvement

2. How does students engage to mathematics subject.

2.1 Liking for learning

2.2 Liking for school

2.3 Effort and persistent

Research Objectives

General Objective: To explore the influence of parental involvement on student engagement in mathematics.

Specific Objectives:

1. To determine the relationship between the level of parental involvement and students' interest and participation in mathematics.

2. To identify the ways in which students engage with the mathematics subject in the presence or absence of parental support.

3. To analyze the impact of different types of parental involvement (e.g., homework assistance, communication with teachers) on students' academic performance and motivation in mathematics.

Hypothesis

This study will be tested at .05 level of significance.

HO: There is no significant relationship between The Influence of Parents Involvement on Students Engagement in Mathematics.

H1: There is a significant relationship between The Influence of Parents Involvement on Students Engagement in Mathematics.

Scope and Delimitations

This study aims to give knowledge on how parental involvement affects the student’s engagement to mathematics subject. The study will only focus on how the parent’s involvements affect the student’s engagement with mathematics and will not

go beyond it. The study will specifically examine how students engage in mathematics subjects with parental involvement.

Theoretical Framework

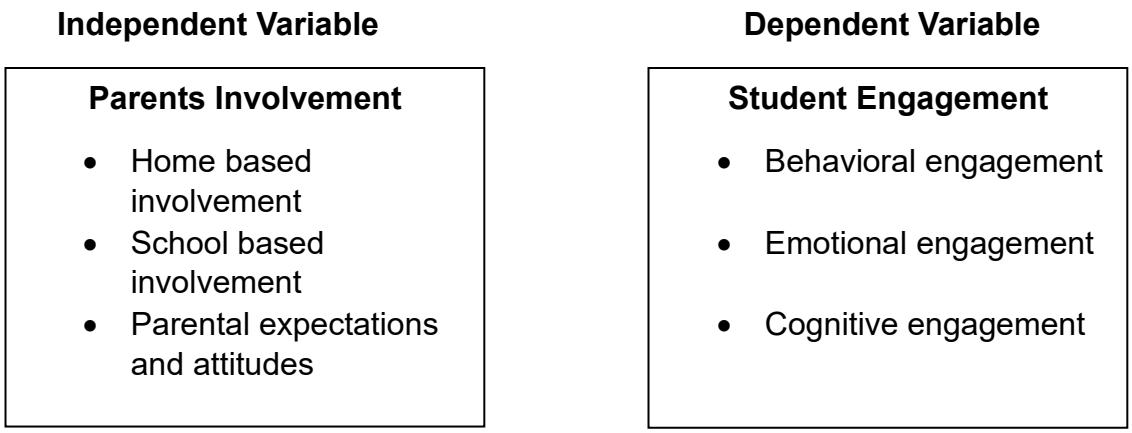
Ecological Systems Theory (Bronfenbrenner, 1979).

This theory posits that a child's development is influenced by various environmental systems, with parental involvement being part of the microsystem, which has a direct and immediate impact on the child's learning. In this context, parental involvement in education, particularly in mathematics, affects students' attitudes, motivation, and engagement in learning. According to Bronfenbrenner, children do better in school, including mathematics, when their parents are actively involved in their education by monitoring homework, attending school events, and creating a conducive learning environment at home.

Expectancy-Value Theory (Eccles & Wigfield, 2002).

This theory suggests that students' motivation and engagement in a subject, such as mathematics, are influenced by their expectations of success and the value they place on the subject. Parental involvement can shape these perceptions by providing support, encouragement, and resources. When parents emphasize the importance of mathematics and communicate high expectations, students are more likely to engage in the subject. The belief that they can succeed in mathematics increases their effort and persistence, which leads to greater engagement.

Conceptual Framework



Significance of the Study

The significance of this study highlights the potential benefits for several key groups:

Teachers – This study will help the teacher to help the students to engage in their classes. Since this study provides insights on how students engage to mathematics with different teaching method.

Students – This study will help students participate in mathematics subject more often, and understand why engaging in mathematics subject helps improve their problem-solving skills.

Parents/Guardians – This study will help the parents/guardians of the students, because they will know that their encouragement will make their student become more active in class.

Future Researchers – The results of this study may help the future researchers as the result of this study may provide them insights, and accurate information. They may

refine and expand studies in relation to parental involvement to students' engagement with the mathematics subject.

Definition of Terms

This part aims to provide a better understanding of the terms used in the study.

Parents Involvement

- Parental involvement refers to the amount of participation a parent has when it comes to schooling and her child's life. Some schools foster healthy parental involvement through events and volunteer opportunities, but sometimes it's up to the parents to involve themselves with their children's education. You can ensure that your child receives the benefits from parental involvements by staying up to date on what is happening in the classroom, helping your child with school opportunities and knowing correct safety procedure for the school.

-operationally described as the active involvement and support of parents in their child's math education, encompassing tasks like offering homework assistance, showing up for school functions, and fostering a love of math education.

-refers to the different ways that parents engage in their child's education—particularly in the area of mathematics. This entails talking with teachers, creating a positive learning atmosphere at home, and inspiring the youngster to do well in math-related tasks.

Students Engagement

- refers to the degree of attention, curiosity, interest, optimism, and passion that students show when they are learning or being taught, which extends to the level of motivation they have to learn and progress in their education.

-Active participation and commitment of students in mathematics, including their attention, interest, effort, and persistence in learning activities.

-refers to the level of attention, interest, and active participation that students exhibit in mathematics, influenced by the involvement and support of their parents in their learning process.

Parental Influence

- Parental influence refers to the significant role that parents play in shaping their children's motivational orientations and behaviors in the context of sport and physical activity.

-The quantifiable effect of parental involvement on the degree of mathematical engagement among students, as evidenced by modifications in their motivation, participation, and performance in math related tasks and evaluation.

-refers to the influence that parental involvement has on how actively students participate and engage in their mathematics studies.

REFERENCES

Nyabuto, A. N., & Njoroge, P. M. (2014). Parental involvement on pupils' performance in mathematics in public primary schools in Kenya. Journal of Educational and Social Research, 4(1), 19-26.

Gutierrez Jr, A. C., & Doronio, R. G. (2024). CLASSROOM SOCIAL ENVIRONMENT AND STUDENT ENGAGEMENT IN MATHEMATICS: A CORRELATIONAL STUDY. European Journal of Education Studies,

Bronfenbrenner, U. (1979). The Ecology of Human Development: Experiments by Nature and Design. Cambridge, MA: Harvard University Press.

Eccles, J. S., & Wigfield, A. (2002). Motivational beliefs, values, and goals. Annual Review of Psychology, 53(1), 109-132.